Korea Technology and Information Promotion Agency for SMEs

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Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

Proposer	**Name**	Hyesoon Kim	**Email**	hyesoon@cc.gatech.edu
Institute Name	Georgia Institute of Technology			
Address	266 Ferst Drive Atlanta GA 30309			

2. Technology Overview

Title of Proposed Technology	System-Aware Compilation for Large-Model Inference on AI Accelerators			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (√), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Information & Communication		Information & Communication	
	Medium Category	Small Category	Medium Category	Small Category
	Software	System Integration	Software	Software Solutions
Concept of Proposed Technology	Large-model inference is increasingly limited by memory capacity and data movement rather than compute. This becomes more pronounced with modern system configurations that include tiered memory and IO, such as host memory expansion with CXL and storage/NVMe in the critical path. The proposed work develops system-aware compilation and system software optimizations that reduce these bottlenecks for inference on AI accelerators, while staying compatible with widely used open-source inference software frameworks. The technical focus is on compiler- and runtime-level optimizations that work within existing open-source serving stacks (for example, vLLM and related systems). Instead of assuming that all tensors			

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	fit neatly in device memory, the system generates execution plans that coordinate compute with movement of data across device memory, host memory, and storage tiers, and that adapt to bandwidth contention.
Rationale for Joint R&D Project	Korea is moving quickly on advanced memory systems, including CXL-based platforms, and many teams are deploying large models under tight cost and memory constraints. At the same time, a large fraction of real deployments depend on open-source inference frameworks (e.g., vLLM-class serving stacks) because they are widely adopted and evolve rapidly. Performance, stability, and cost in these deployments are often determined by data movement, not by raw accelerator computing units. This joint R&D is needed to develop and validate system-aware compilation and system software optimizations directly in the context of these widely used open-source frameworks. It focuses on optimizations that can be upstreamed or cleanly carried as optional modules, and that are validated on representative workloads and configurations (including tiered memory such as CXL and storage-heavy pipelines).
Key Words	1. System-aware compilation 2. AI accelerator inference (NPU/accelerator) 3.CXL memory expansion and tiered/disaggregated memory 4. Compiler

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL) [2]	Start of the Project	Level (2)	End of the Project	Level (5)
R&D Objectives	The objective is to develop deployable compiler and system software optimizations that improve large-model inference efficiency in memory- and IO-constrained settings, with implementation and evaluation centered on open-source inference frameworks. Specific objectives: ● develop memory planning and buffer orchestration to reduce peak device memory and support larger-than-device-memory inference ● develop streaming execution across device/host/CXL/storage tiers with overlap of compute and transfers ● develop contention- and IO-aware scheduling policies to improve throughput stability and reduce tail latency ● integrate and evaluate the techniques in open-source inference frameworks (e.g., vLLM-class stacks) with reproducible benchmarking			
R&D Target Specifications	throughput improvement in memory/IO-bound settings: 1.2×–2.0× on representative pipelines practical deployability: demonstrated in at least one open-source inference framework with reproducible evaluation artifacts			
R&D Plan	Year 1: baseline and evaluation system setup ● collect partner workloads and constraints (model shapes, batch patterns, latency SLOs, memory limits, IO topology) ● establish baselines in a chosen open-source inference framework and define benchmark suite ● implement memory planning and buffer orchestration prototypes; evaluate memory reduction and overhead Year 2: streaming execution and contention-aware scheduling ● add streaming execution and overlap of compute/transfer across memory/storage tiers ● implement contention/IO-aware scheduling and validate under bandwidth variability and multi-stream workloads ● harden integration points so optimizations can be toggled and evaluated cleanly			

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	Year 3: profiling-guided policies and pilot validation ● add profiling-guided policy selection to reduce manual tuning across systems ● pilot-scale validation on partner-relevant configurations (including tiered memory such as CXL and storage-heavy settings) ● package reproducible scripts and documentation; prepare a technology transfer package suitable for adoption

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	- design and implement compiler and system software optimizations (memory planning, buffering, streaming execution, scheduling, profiling-guided policies) - Integrate prototypes into selected open-source inference frameworks and maintain reproducible evaluation harnesses - lead benchmarking, analysis, and reporting; support joint debugging and iteration - Deliver reference implementations and documentation focused on the optimization techniques and integration interfaces
Expected R&R of Korean SME Partner	- provide representative inference pipelines, deployment constraints, and success metrics (latency/throughput/memory footprint; power) - provide validation environments and operational insights - participate in testing and pilot demonstrations; provide feedback on deployability and maintenance burden - contribute to downstream adoption planning

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	This joint R&D is expected to deliver practical compiler and system software optimizations that reduce memory pressure and data-movement overhead for large-model inference deployments. By targeting widely used open-source inference frameworks, the results can be validated quickly and transferred in a form that is easy to adopt. Expected outcomes include: ● improved deployability of large models when device memory is limited, including tiered-memory and storage-heavy scenarios ● improved throughput stability and reduced tail latency in IO-coupled inference pipelines ● reference implementations integrated with an open-source inference framework and accompanied by reproducible benchmarks ● stronger competitiveness for Korean SMEs that deploy large-model inference services on modern memory systems (including CXL-based configurations)

6. References

(1) Jiashen Cao, Joy Arulraj, Hyesoon Kim, Buffer Management for Out-of-GPU LLM Execution, SIGMOD DEEM Workshop, 2025.

(2) Xueyang Liu, Seonjin Na, Euijun Chung, Jiashen Cao, Jing Yang, Hyesoon Kim, Contention-Aware GPU Thread Block Scheduler for Efficient GPU-SSD, IEEE Computer Architecture Letters (CAL), 2025.

(3) Seonjin Na et al., FlexInfer: Flexible LLM Inference with CPU Computation, MLSys, 2025.

(4) Shinnung Jeong et al., Inside VOLT: Designing an Open-Source GPU Compiler, CC, 2026

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization